

Article

Design, Construction, and Simulation-Based Validation of a High-Efficiency Electric Powertrain for a Shell Eco-Marathon Urban Concept Vehicle

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Abstract

This study addresses a documented gap in detailed, cost-effective, and performance-validated electric vehicle (EV) powertrain solutions. It presents the complete design, construction, and simulation-based validation of a high-efficiency electric powertrain for a Shell Eco-marathon Urban Concept vehicle. Novel contributions of this work include a unique drivetrain architecture: a BLDC motor with a modular two-stage chain drive and a custom lithium-ion battery pack. The design is optimized for compactness and reliability under stringent budget and packaging constraints. A comprehensive Simulink-based vehicle dynamics model was developed for robust validation. This model enabled the estimation of energy consumption, torque profiles, and battery State of Charge under realistic drive cycles. The system demonstrated a remarkably low energy consumption under competition conditions, signifying high efficiency with <50 Wh/km consumption and full compliance with technical regulations. Furthermore, the hardware is thoroughly documented with detailed build instructions, CAD models, and a full bill of materials. This promotes reproducibility. This research offers a validated, low-cost, and replicable electric powertrain. It provides a transferable framework for future Shell Eco-marathon teams and advances lightweight, cost-effective solutions for real-world low-speed electric mobility applications, such as micro-EVs and urban delivery vehicles.

Keywords: electric vehicle; powertrain design; Shell Eco-marathon; BLDC motor; Simulink; energy efficiency; drivetrain; battery pack



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1. Introduction

The global shift toward electric mobility is accelerating, driven by the urgency to reduce greenhouse gas emissions, improve urban air quality, and lessen reliance on fossil fuels. According to the International Energy Agency (IEA), electric vehicles (EVs) are expected to comprise more than 30% of global vehicle sales by 2030 [1]. The Shell Eco-marathon (SEM) competition provides a unique academic platform where student teams design and build energy-efficient vehicles under stringent technical, financial, and regulatory constraints.

The Urban Concept (UC) category replicates real-world micro-mobility challenges, including enclosed bodywork, stop-and-go driving, and packaging limitations [2]. Coventry University's BA Momentum team developed a 48 V electric UC vehicle with a limited

budget (GBP 13,500) and strict packaging constraints, mirroring trade-offs encountered in commercial micro-EV applications. Despite a surge in EV research, detailed, validated, and low-cost drivetrain architectures suited for academic and competition contexts remain under-documented. This study addresses that gap by presenting the design, construction, and simulation-based validation of a high-efficiency electric powertrain for SEM UC requirements.

The work follows a systems engineering approach, incorporating structured decision matrices for component selection, custom lithium-ion (Li-ion) battery design, and validation via MATLAB/Simulink. The drivetrain integrates a brushless direct current (BLDC) motor with a modular two-stage chain drive and custom battery pack. The contributions include open-access CAD (computer-aided design) models, a full bill of materials (BoM), and validated simulation results offering a reproducible framework for SEM teams and micro-EV developers.

Research Gap and Motivation

While previous studies address EV component selection and lightweight chassis design [3,4], few present an integrated drivetrain solution with a rigorous methodology, simulation-based validation, and competition-ready packaging. Moreover, low-voltage, cost-effective drivetrain designs remain underreported in the academic literature. This project therefore emphasizes modularity, energy efficiency, and compliance with SEM regulations, bridging a critical knowledge gap.

Motor selection is one of the most critical factors influencing drivetrain performance, as shown in Figure 1. Motor choice is central to drivetrain performance. Comparative studies have evaluated brushed DC (BDC) motors, induction motors (IMs), switched reluctance motors (SRMs), permanent magnet synchronous motors (PMSMs), and BLDC motors [4,5].

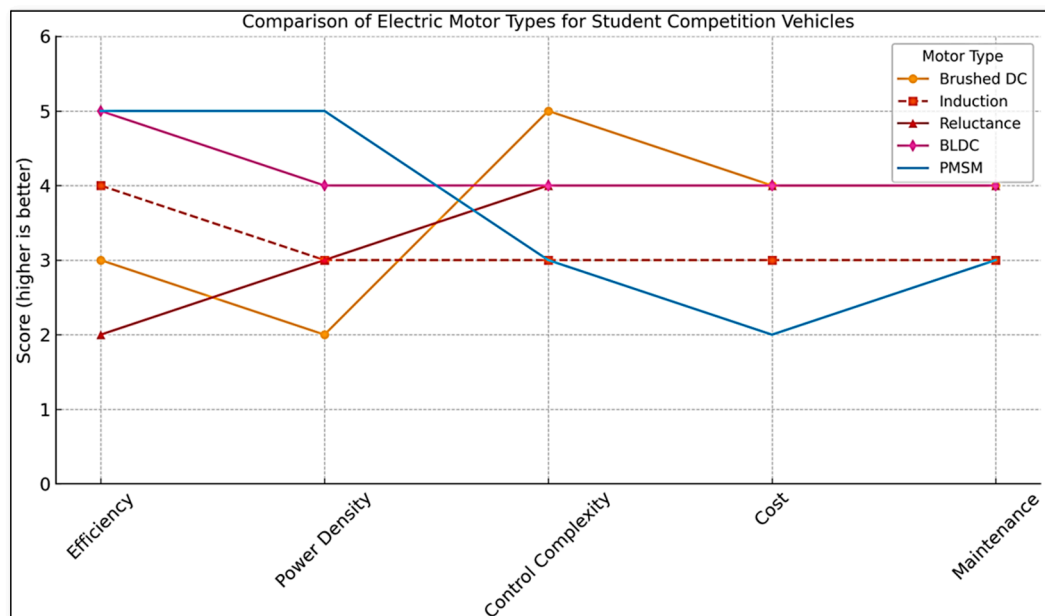


Figure 1. Comparison of electric motor types on key performance criteria (adapted from [4]).

BLDC motors consistently achieve >90% efficiency, strong reliability, and manageable control complexity, making them the preferred option for student-built competition vehicles. PMSMs and axial flux motors (AFMs) offer higher performance but at significantly greater cost and manufacturing complexity [6–8]. Thus, a BLDC motor was selected as the optimal compromise between efficiency, cost, and simplicity.

Powertrain architecture also impacts efficiency and maintainability. Advanced options like dual in-wheel motors (IWMs) provide precise torque control and regenerative braking (RB) [9–12] but add unsprung mass (UM), weight, and complexity. Most SEM teams favour single central motors for simplicity and reliability, which aligns with the present design choice.

Transmission type is another determinant of efficiency. Gear drives (GDs) achieve >99% efficiency but are difficult to prototype affordably [13]. Chain drives, widely used in Shell Eco-marathon vehicles, offer mechanical efficiency of around 98% and allow for easy modification of gear ratios [14]. Adaptive chain tensioning, as investigated by [15], can further enhance drivetrain efficiency under variable loads. Hybrid transmission designs, such as those combining belt and gear stages, are emerging as viable alternatives for micro-mobility applications [16]. Ref. [17] experimentally demonstrated that doubling sprocket ratios increased chain efficiency by 2–5%, and quadrupling chain tension improved it by 18%. Meanwhile, belt systems showed 34.6% higher frictional losses compared to chain drives under an equivalent preload [18]. A two-stage CD was therefore selected as the best trade-off between performance, serviceability, and compliance.

Multi-stage transmissions offer a promising middle ground. The PETRONAS University of Technology team addressed chain breakage issues by introducing a hybrid drivetrain with a belt stage for high-torque loads followed by a chain stage for moderate-speed transmission [19]. Their design reduced strain on the chain and allowed better durability under dynamic loads. Ref. [15] also showed how adaptive chain tensioning can further improve drivetrain efficiency under variable loading. It was mentioned that the vehicle equipped with a chain drive system from the previous year suffered from chain breakage, as the chain was connected directly to the engine output shaft and transmission, and the direct load from the engine exceeded the load capacity of the chain. To address the issue, the team designed a two-stage transmission—the first stage was a belt system which had the ability to sustain the high rotation speed and torque directly from the engine, whereas the second stage was a chain system that was more reliable at a lower rotation speed and torque. A diagram of their design is given in Figure 2.

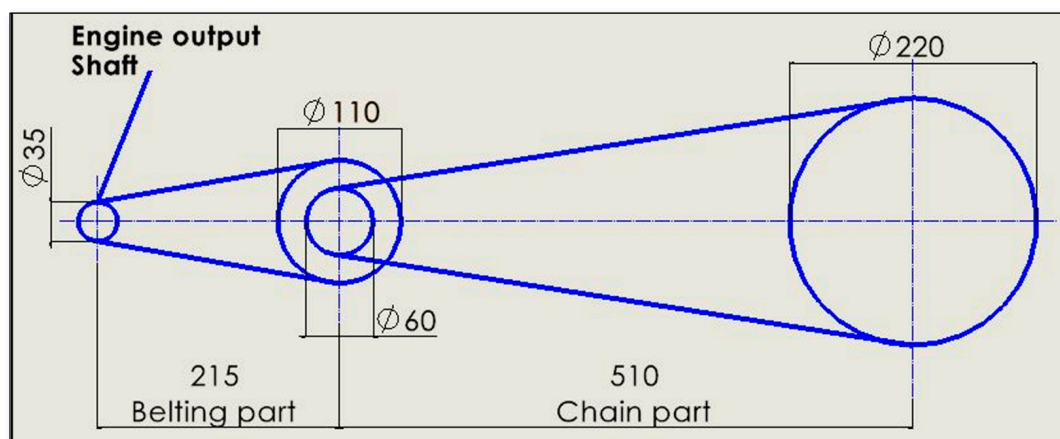


Figure 2. Two-stage transmission designed at PETRONAS University of Technology (adapted from [19]).

In a study by Spicer et al. [17], frictional losses in a bicycle chain drive were analysed through experimentally measuring its efficiency. It was found that the chain drive efficiency increased with the tooth ratio and chain tension. Results from their experiments showed a 2–5% increase in efficiency when doubling the sprocket ratio, whilst quadrupling the chain tension raised the efficiency by 18%. The maximum efficiency recorded was 98.6%. Even

though the belt was less efficient than the chain with low load application because of the high pretension required in the belt, it was found that the belt could have less frictional losses if the pretension remained under 40 lbf.

Li-ion cells, such as Molicel P28A, dominate SEM applications due to their energy density (ED) and discharge characteristics [20,21]. Accurate modelling of State of Charge (SoC), internal resistance (IR), and thermal behaviour (TB) is critical for predicting range and energy use [22].

MATLAB/Simulink R2022a frameworks have been shown to reliably replicate competition drive cycles (DCs) within 5% accuracy [23–25]. This supports the present study's approach of integrating simulation-driven validation with experimental design.

The recent literature supports the use of BLDC motors, chain-driven single-motor configurations, and modular lithium-ion battery systems for Shell Eco-marathon vehicles and related applications. The emphasis on system integration, modular simulation, and component-level validation reflects a maturing field that balances academic rigor with real-world constraints. While more advanced technologies such as axial flux motors, adaptive transmissions, or real-time thermal modelling show promise, they remain out of reach for many student-led teams due to budget and complexity. The current study builds on these findings, implementing a validated, cost-effective drivetrain architecture optimized for Shell Eco-marathon constraints. Its methodology, grounded in recent research and supported by simulation, offers a transferable framework for future developments in low-speed electric mobility, including last-mile delivery vehicles and urban micro-EVs.

Relevant studies support the growing interest in energy-efficient EV design for constrained applications. For example, ref. [26] presents a complete chassis design and FEA validation for a Shell Eco-marathon vehicle, focusing on lightweight structural optimization, a complementary domain to our drivetrain-oriented approach. Meanwhile, ref. [23] explores inductive power transfer for IoT-level electric vehicles, which highlights future trends in wireless charging but lies outside the current scope of low-cost, competition-focused designs. Compared to these works, the present study uniquely contributes a fully documented, simulation-validated powertrain architecture optimized for both efficiency and replicability under Shell Eco-marathon constraints.

In addition to competition-focused studies, several open-source EV platforms such as OSVehicle's Tabby EVO [27] have emerged in recent years to support modular, low-cost electric vehicle development. These platforms promote open-access prototyping, yet most lack simulation-based validation or integration with regulated competition environments. Recent research supports BLDC motors, chain-driven single-motor setups, and modular Li-ion battery systems as optimal solutions for SEM vehicles. While advanced technologies such as hybrid transmissions (HTs) and wireless charging (WC) hold promise, they remain impractical for budget-constrained student projects. The present study contributes a validated, low-cost drivetrain framework optimized for SEM conditions and scalable to real-world low-speed EVs, including urban delivery and micro-mobility platforms [28–31].

2. Powertrain Design

This section presents the full design process of the Shell Eco-marathon Urban Concept vehicle's electric powertrain, covering initial vehicle calculation, system architecture, component selection, drivetrain configuration, and gear ratio optimisation. The focus is on efficiency, reproducibility, and compliance with competition rules.

2.1. Initial Vehicle Calculations

Traction Force Calculation

Vehicle performance requirements were estimated using the assumptions listed in Table 1.

Table 1. Input parameters.

Parameters	Value	Unit
Vehicle mass including driver (m)	295	kg
Wheel diameter (d)	0.5715	m
Distance (s)	16	km
Duration (t)	40	min
Acceleration time (t_a)	20	s
Rolling coefficient (C_r)	0.015	
Drag coefficient (C_d)	0.3	
Frontal area (A)	1.69	m ²
Gravity (g)	9.81	m/s ²
Air density (ρ)	1.225	kg/m ³

The average speed of the vehicle is calculated as follows:

$$\text{Average speed} = \frac{s}{t} = \frac{16 \times 1000}{40 \times 60} = 6.944 \text{ m/s}$$

By dividing the average speed by the acceleration time, the acceleration of the vehicle is

$$\text{Acceleration} = \frac{\text{Average speed}}{t_a} = \frac{6.944}{20} = 0.347 \text{ m/s}^2$$

The total traction force was calculated as the sum of acceleration, rolling, and aerodynamic drag components:

$$\text{Acceleration force(N)} = m \times \text{acceleration} = 295 \times 0.347 = 102.43 \text{ N}$$

$$\text{Rolling force(N)} = m \times g \times C_r = 295 \times 9.81 \times 0.015 = 43.375 \text{ N}$$

$$\text{Drag force(N)} = m \times g \times C_d = 295 \times 9.81 \times 0.3 = 14.976 \text{ N}$$

$$\text{Total traction force(N)} = 102.43 + 43.375 + 14.976 = 160.78 \text{ N}$$

The corresponding wheel torque, power demand, and total energy consumption over the race duration were obtained using the following:

$$\text{Torque(N)} = \text{Total traction force} \times \frac{d}{2} = 160.78 \times \frac{0.5715}{2} = 22.971 \text{ Nm}$$

$$\text{Power(W)} = \text{Total traction force} \times \text{Average speed} = 160.78 \times 6.944 = 1116.521 \text{ W}$$

$$\text{Energy(Wh)} = \text{Power} \times t = 1116.521 \times \frac{40 \times 60}{3600} = 744.347 \text{ Wh}$$

$$\text{Energy consumption} = \frac{\text{Energy}}{s} = \frac{744.347}{16} = 46.522 \text{ Wh/km}$$

From this, the required battery rating was calculated. A 10% cutoff in battery discharge was employed to ensure the health of the battery.

$$\text{Battery rating(Wh)} = \frac{\text{Energy}}{0.9} = \frac{744.347}{0.9} = 827.052 \text{ Wh}$$

All data calculated are presented as a summary in Table 2.

Table 2. Output of calculations.

Parameters	Value	Unit
Average speed (v)	6.95	m/s
Acceleration (a)	0.35	m/s ²
Total traction force (F _t)	160.78	N
Torque (T)	22.97	Nm
Power (P)	1116.52	W
Energy (E)	744.35	Wh
Energy consumption	46.52	Wh/km
Battery rating	827.05	Wh

2.2. Design Overview

2.2.1. High-Level Architecture

The electric powertrain developed for the Shell Eco-marathon Urban Concept vehicle comprises four key components, designed to maximize efficiency, reliability, and maintainability within strict regulatory and budgetary constraints:

- **Brushless DC (BLDC) Motor:** A 1500 W, 48 V BLDC motor (HPM-1500B, Golden Motor Co., Guangzhou, China) was chosen due to its high efficiency (~85%), strong power-to-weight ratio (5.5 kg), and minimal maintenance requirements. Mounted centrally, it delivers consistent propulsion and integrates seamlessly with the chassis design.
- **Two-Stage Chain Drive Transmission:** This modular system enables a high gear ratio (12:1) and a medium gear ratio (8:1) using compact sprockets, ideal for varied speed targets (e.g., 25 km/h and 40 km/h). It efficiently transfers power from the motor to the rear axle while minimizing mechanical losses and addressing spatial constraints within the chassis.
- **Battery Pack:** A custom 12S8P lithium-ion battery pack, built from Molicel P28A 18650 cells(Molicel, Taipei, Taiwan), serves as the energy source. It delivers a nominal 43.2 V and approximately 967 Wh—within the Shell Eco-marathon’s regulatory limit of 1000 Wh and 60 V.
- **Motor Controller + DC-DC Converter:** A programmable controller governs motor torque based on PID-calculated torque demand, while a Victron Orion-Tr 48/12-9A DC–DC converter (Victron Energy, Almere, The Netherlands) powers auxiliary systems. Integrated current, voltage, and State of Charge (SoC) monitoring ensures safe and efficient energy management throughout the drive cycle.

This architecture offers a robust combination of high performance, modularity, and cost-effectiveness, tailored for ultra-efficient vehicle competitions and real-world micro-mobility applications.

This configuration was validated using a MATLAB Simulink R2022a (The MathWorks Inc., Natick, MA, USA). The results confirmed the energy consumption previously reported, with high-fidelity speed tracking and an effective torque response across the Shell Eco-marathon drive cycle.

2.2.2. Block Diagram of the Full Powertrain System

The goal of the simulation was to model the energy consumption and driving performance of an electric powertrain vehicle using the selected BLDC motor rated at 4.78 Nm, 3000 rpm, and 39 A peak current, paired with a designed battery pack from last year's powertrain team with a custom lithium-ion battery pack. The model was created entirely using basic Simulink blocks, as shown in Figure 3.

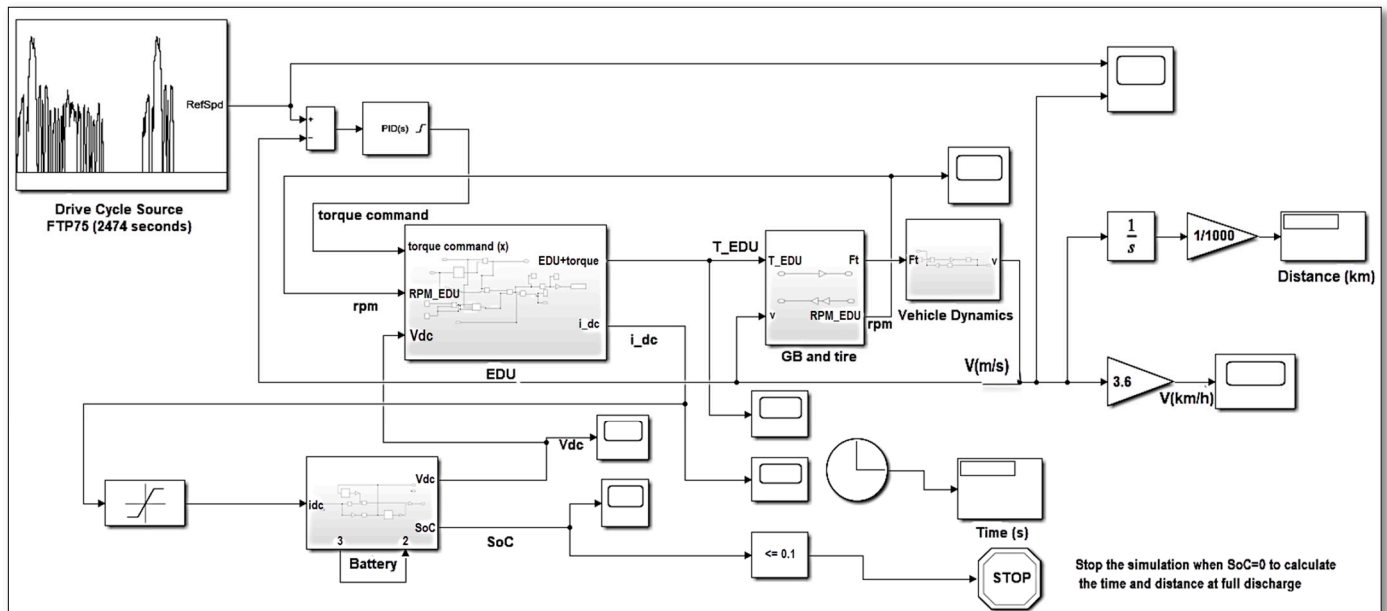


Figure 3. Simulink model of Shell Eco-marathon 2025 prototype electric vehicle.

The simulation structure follows a modular architecture:

- **Drive Cycle Source:** This includes a 293 s drive cycle map with the vehicle's target speed over time based on the Shell Eco-marathon track.
- **Speed Control Loop:** A PID controller compares the reference and actual speed to generate the torque demand.
- **Electric Drive Unit (EDU):** This converts voltage and torque input into shaft speed and current drawing based on the motor's performance characteristics.
- **Battery Model:** This simulates the voltage and State of Charge (SoC) dynamics using the input current and internal resistance logic.
- **Vehicle Dynamics Block:** This transfers motor torque into forward motion, acting in simplified drivetrain losses and rolling resistance.
- **Distance and State of Charge (SoC) Tracking:** The total distance travelled and SoC are calculated over time.
- This approach will allow estimation of how long the vehicle can run and how far it can travel on a full battery, under the idealized drive cycle.

Material selection was based on efficiency. The selected BLDC motor is capable of delivering 4.78 Nm at 3000 rpm, with a current draw of up to 39 A. The power source consisted of the custom lithium-ion battery pack, selected for its high discharge rate (up to 35 A) and stable voltage underload. A simple 12 series, 8 parallel battery model was used, estimating the voltage and State of Charge (SoC) from the current drawn using the energy balance equations.

The simulation also includes safety requirements by ensuring that the discharge current and voltage remained consistent with the Shell Eco-marathon 2025 official rules.

2.3. Motor and Transmission Selection

2.3.1. Motor Selection Justification

The selected BLDC motor, shown in Figure 4, offers a compelling balance of efficiency, power density, and control simplicity. As highlighted by Monteiro et al. [4], BLDC motors outperform induction, brushed DC, and reluctance motors across multiple criteria: efficiency (>90%), weight, size, and maintenance. The BLDC motor was selected as the power unit, offering a strong balance of efficiency, compactness, and ease of control. Compared with induction, brushed DC, and reluctance motors, the chosen BLDC achieved 85% efficiency at a weight of 5.5 kg, making it well-suited to the Urban Concept platform.



Figure 4. Three-dimensional view of selected motor.

Its technical specifications are listed below:

- Rated power = 1500 W;
- Rated voltage = 48 V;
- Rated current = 39.06 A;
- Rated torque = 4.78 Nm;
- Rated speed = 3000 rpm;
- Efficiency = 85%.

The speed and torque of the motor when used at its maximum efficiency point are 2991 rpm and 2.91 Nm, while at its maximum torque point, the speed and torque are 2944 rpm and 5.15 Nm, as shown in Table 3.

Table 3. Selected motor specifications.

Parameters	Speed (rpm)	Torque (Nm)
Maximum efficiency point	2991	2.91
Maximum torque point	2944	5.15

2.3.2. Gear Ratio Calculation

Considering that the vehicle should be able to run at its highest efficiency at two different speeds, 25 km/h for the Super mileage challenge and 40 km/h for the Shell Eco-marathon regional championship, two gear ratios were needed to achieve this.

The equation below shows the relationship between the gear ratio, speed, and torque. It was seen that the gear ratio (represented by the ratio between the diameter d of driven and driving gears) was positively proportional to torque (T) and inversely proportional to speed (n). Therefore, when the gear ratio was raised, the torque increased and speed decreased.

$$\frac{d_2}{d_1} = \frac{n_1}{n_2} = \frac{T_2}{T_1}$$

Based on the motor speed and torque during different motor operations listed in Table 4, the resulting speed and torque at the wheels at different gear ratios were calculated. It was found that gear ratios of 12 and 8 allowed the motor to be used most efficiently when cruising at 25 km/h and 40 km/h, respectively.

Table 4. Gear ratios and their corresponding calculated outputs.

Gear Ratio	12	8
Most efficient vehicle speed (km/h)	26.9	40.29
Most efficient wheel torque (Nm)	34.9	23.3
Maximum vehicle speed (km/h)	33.6	50.4
Maximum wheel torque (Nm)	61.8	41.2

Regarding drivetrain layout, the torque and power transmission from the motor to the wheels is considered in this section. Three common types of drivetrains are compared, gear, chain, and belt, as shown in Figure 5.

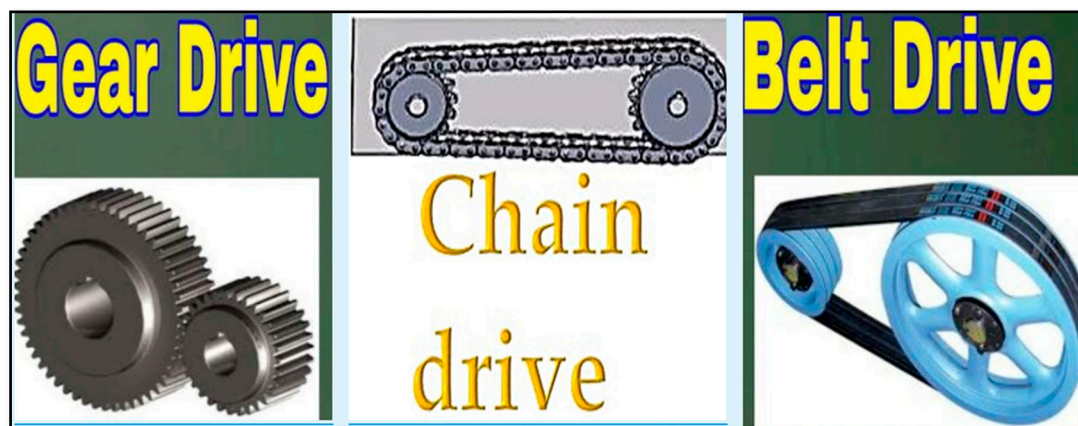


Figure 5. Several types of drivetrain layouts (adapted from Childs, 2018).

The drivetrain layout is evaluated in terms of torque and power transmission from the motor to the wheels. Gear drives generally achieve the highest efficiency (>99%) but are costly and difficult to prototype; chain drives offer high efficiency (~98%) with easy ratio modification, making them common in Shell Eco-marathon vehicles; while belt drives are simpler to maintain but suffer from higher frictional losses.

2.3.3. Comparative Decision

As stated in the Shell Eco-marathon rules, a maximum of two motors are allowed to be employed in the vehicle. The team proposed three potential motor layouts: a single motor

with transmission, independent dual motors, and dual in-wheel motors. The single-motor configuration is commonly adapted in internal combustion vehicles with a single power supply from the engine: power is transmitted to the wheels through the transmission and differential. Independent dual motors have the benefit of multiplying torque through gear ratios while enhancing the control of wheels by driving wheels independently; hence, there is no need for a differential in this configuration. Dual in-wheel motors are an advanced technology; the motors are assembled inside the wheel hub and drive the wheels directly without transmission to achieve the ultimate control of wheels.

Motor(s) with a total rating of 1500 W would be suitable for producing the required power, i.e., incorporating a 1500 W motor in a single-motor layout or two 750 W motors in a dual-motor layout. For comparison, the cost and weight of brushless motors of the mentioned ratings were taken as an example. From Table 5, the estimated cost and weight of the motors alone in a dual-motor configuration would be GBP 545.5 and 9 kg, compared to GBP 334.29 and 5.5 kg for a single-motor configuration. In-wheel motors are believed to be more expensive than conventional electric motors due to the complex integration parts and high torque output.

Table 5. Comparison of motors.

Brushless Motors		
Motor	1500 W 48 V	750 W 48 V
Cost	GBP 334.29	GBP 272.75
Weight	5.5 kg	4.5 kg

The single-motor configuration was selected for its superior cost-efficiency, simplicity, and ease of integration with the mechanical layout. While dual and in-wheel setups offered certain advantages, their added weight, control complexity, and cost disqualified them under current project goals. This methodical selection process ensured that the powertrain achieves top performance while staying compliant with Shell Eco-marathon technical regulations and within budget. Finally, three motor layouts (single motor with transmission, dual independent motors, and in-wheel motors) were evaluated.

2.4. Battery Pack Design

2.4.1. Cell Layout

To power the Shell Eco-marathon Urban Concept vehicle within the strict competition constraints, the team designed a custom lithium-ion battery pack using Molicel P28A 18650 cells. The adopted layout is a 12S8P configuration, meaning 12 cells in series and 8 in parallel, for a total of 96 cells. The battery pack designed last year consisted of 96 Molicel P28A cells (Molicel, Taipei, Taiwan) in a layout of 12S8P. The specifications of the design are listed in Table 6.

Table 6. Battery details.

Parameter	Value
Nominal pack voltage (V)	43.2
Total energy (W h)	967.68

The battery pack was designed using 96 Molicel P28A cells (12S8P), delivering 43.2 V and 22.4 Ah (967.7 Wh), which is within the Shell Eco-marathon 1000 Wh limit. Each cell offers 3.6 V, 2.8 Ah, and a 35 A discharge rate, ensuring stable voltage and high energy

density. Simulink validation showed a range of 20.9 km (equal to 41 min at 30 km/h). The pack weighs 8.5 kg, costs around GBP 860, and integrates a BMS and charger.

2.4.2. Electrical Specifications

To meet the strict constraints of the Shell Eco-marathon competition, the team developed a custom lithium-ion battery pack based on 96 Molicel P28A 18650 cells arranged in a 12S8P configuration, providing a nominal voltage of 43.2 V and a total energy capacity of 967.68 Wh. This design falls within the SEM electrical limits of 60 V and 1000 Wh.

The cells were selected for their high discharge capability (up to 35 A), stable voltage characteristics, and competitive energy density. A Battery Management System (BMS) was integrated into the battery enclosure to ensure operational safety through cell balancing and overvoltage, undervoltage, and overcurrent protection, in accordance with SEM 2025 rules.

Simulation of the pack within the complete vehicle model demonstrated effective energy delivery, with a single-lap energy consumption of 20.95 Wh and a remaining State of Charge (SoC) of 97.77%, verifying both safety and sufficiency. The system voltage and current remained within safe operational thresholds under competition drive cycles.

All detailed electrical parameters, simulation outputs (voltage/current profiles, SoC), and CAD renderings of the pack design have been moved to the Supplementary Materials to support reproducibility while maintaining conciseness in the main manuscript.

2.5. Structural Integration

The structural integration of the powertrain components, including the motor, drive-train, and battery pack, was performed to ensure a compact, manufacturable, and serviceable system that meets Shell Eco-marathon rules and packaging constraints of the Urban Concept chassis.

CAD Models and Chassis Constraints

The first concept design was constructed based on all the decisions made in the earlier sections. As shown in the schematic diagram in Figure 6, the powertrain consisted of a single motor that drives the rear wheels through a single-stage chain drive and a differential.

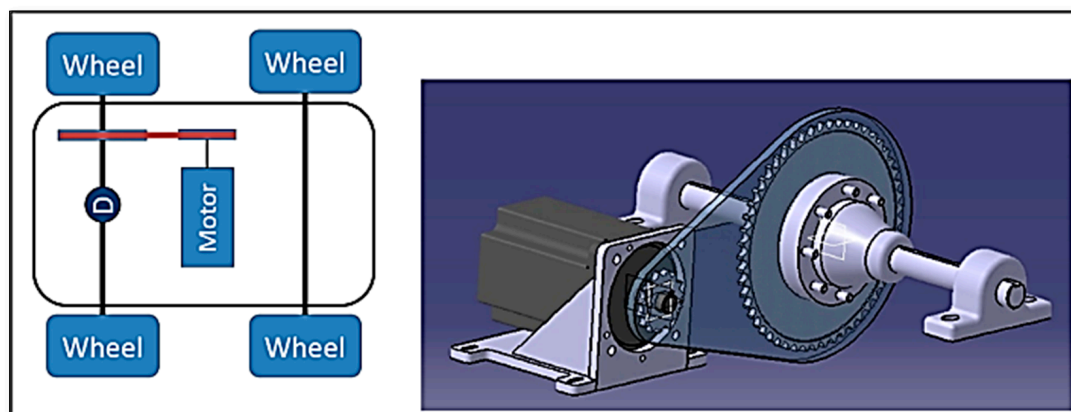


Figure 6. Overview of single-motor design.

The size of the sprocket is defined by the number of teeth, and to achieve a certain gear ratio, the number of teeth on the larger sprocket (on the axle) should be the product of the gear ratio and the number of teeth on the smaller sprocket (on the motor). Hence, when deciding on the sprocket size, the smaller sprocket should be as small as possible to avoid the larger sprocket taking up a larger amount of space.

Considering that the gear ratio of 12 was quite large, the smallest sprocket size of 10 teeth was chosen for the small sprocket at the motor. The size of other sprockets is listed in Table 7.

Table 7. Sprocket details to provide the required gear ratios.

Sprocket	Number of Teeth (Gear Ratio = 12)	Number of Teeth (Gear Ratio = 8)
Motor (small)	10	10
Axle (large)	120	80

Extensive CAD modelling was performed using CATIA V5 (Dassault Systèmes, Vélizy-Villacoublay, France) to simulate the full assembly of the drivetrain and battery system within the vehicle's rear chassis. As shown in Figure 7, the powertrain layout was validated in 3D to ensure compatibility with space limitations and mounting surfaces. A list of parts is shown in Table 8.

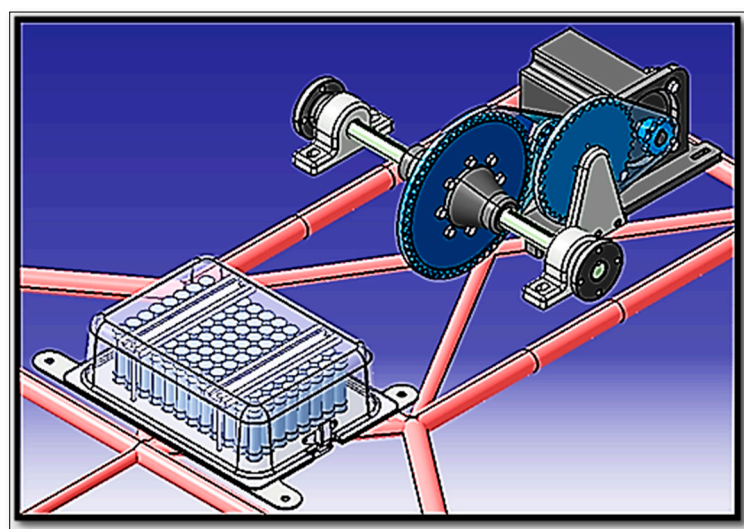


Figure 7. Full EV assembly in context.

Table 8. Bill of materials.

Quantity	Part	Material Type	Product Code	Source/Supplier	Cost (GBP)
1	1.5 kW brushless motor	Steel, copper, permanent magnets	Golden Motor HPM-1500B, 48 V, 3000 rpm	Buy OTS, Golden Motor Co. Guangzhou, China https://www.goldenmotor.com , accessed on 1 September 2025	450
1	Upright motor mount	Aluminium 6061-T6	Custom CAD part	In-house fabrication	0
1	Mount base	Aluminium 6061-T6	Custom CAD part	In-house fabrication	0
2	Rib	Aluminium 6061-T6	Custom CAD part	In-house fabrication	0
2	Side piece	Aluminium 6061-T6	Custom CAD part	In-house fabrication	0
2	12T sprocket	Steel	428-12T sprocket	Buy OTS, JT Sprockets, Wuppertal, Germany https://www.jtsprockets.com , accessed on 1 September 2025, (JT Sprockets, Wuppertal, Germany)	46.74
1	36T sprocket	Steel	428-36T sprocket	Outsource then change	50
1	Gear connector	Aluminium 6061-T6	Custom CAD part	Manufacture in house	0

Table 8. Cont.

Quantity	Part	Material Type	Product Code	Source/Supplier	Cost (GBP)
4	5 mm key	Steel	Parallel key 5 mm × 5 mm × 20 mm (DIN 6885)	Buy OTS, RS Components, Corby, UK https://uk.rs-online.com , accessed on 1 September 2025	20
2	Bearing	Steel (with lubricant)	W6282Z Budget Stainless Steel Metal Shielded Deep Groove Ball Bearing 8 × 24 × 8 mm	Buy OTS https://simplybearings.co.uk/shop/p152973/W6282Z-Budget-Stainless-Steel-Metal-Shielded-Deep-Groove-Ball-Bearing-8x24x8mm/product_info.html , accessed on 1 September 2025 (Simply Bearings Ltd., Leigh, UK)	6.72
1	8 mm silver steel rod	Steel	Silver steel Ø8 mm, grade BS1407	In-house machining	0
2	428 chain	Steel	DID 428H Roller Chain, 120 links	Buy OTS, STANDARD Chains—RK EUROPE, Heerlen, The Netherlands https://rk-europe.com/chain/standard-chains/ , accessed on 1 September 2025 (RK EUROPE, Heerlen, The Netherlands)	68
2	Axle bearings and housings	Steel/aluminium	UCFL205 Flanged Bearing Housing	Buy OTS, SKF/Simply Bearings, https://simplybearings.co.uk/shop/ https://simplybearings.co.uk , accessed on 1 September 2025 (SKF, Gothenburg, Sweden)	73.38
1	Peerless 110 differential and axle	Steel/alloy	Peerless 110 Series Differential and Axle Assembly	Buy OTS then Modify Peerless 110 Series HD Differential—Gemini Karts, Northampton, UK https://www.geminikarts.co.uk/product/peerless-110-series-hd-differential/ , accessed on 1 September 2025 (Peerless Gear, Salem, IN, USA)	195
1	48T sprocket	Steel	428-48T sprocket	Outsource then change, JT Sprockets	50
2	Half-shaft mounts	Aluminium 6061-T6	Custom CAD part	In-house fabrication	0
96	Battery cells	Lithium ion (18650, Molicel P28A)	Molicel P28A 18650, 3.6 V, 2.8 Ah	Buy OTS, Nkon B.V. Eindhoven, The Netherlands https://eu.nkon.nl , accessed on 1 September 2025	312.96
1	Battery connectors/bus bars	Copper	Copper bus bars, 5 mm × 20 mm	Buy OTS, RS Components	332
1	Wiring	Copper with PVC insulation	Silicone insulated wire, 10 AWG	Buy OTS, RS/Farnell	30
1	BMS	Mixed electronic components	ANT BMS 12S Li-ion, 60 A, UART/BT	Buy OTS, AliExpress/Daly BMS Store	40
1	Battery box cover	Plastic (ABS)	Custom vacuum-formed ABS, 3 mm	In-house fabrication	0
1	Battery plate	Aluminium 6061-T6	Custom CAD part	In-house fabrication	0
2	Box clips	Steel	Toggle latch, 50 mm	Buy OTS, Amazon/RS	10
1	DC/DC converter	Mixed electronic components	Victron Orion-Tr 48/12-9A Isolated	Buy OTS https://www.sunshinesolar.co.uk/ , accessed on 1 September 2025	57
1	Charging components	Mixed electronic components	Li-ion Charger 48 V, CC/CV mode	Buy OTS, Nkon/AliExpress	28.49

Table 8. *Cont.*

Quantity	Part	Material Type	Product Code	Source/Supplier	Cost (GBP)
2	Motor controller	Mixed electronic components	Sabvoton SVMC72150 FOC Controller (48–72 V)	Buy OTS, Sabvoton/AliExpress	200
1	Motor control unit	Mixed electronic components	VCU (Vehicle Control Unit), STM32-based custom PCB	Buy OTS In-house design + JLCPCB https://jlcpcb.com , accessed on 1 September 2025	100

During the initial design, the use of a large 120-tooth sprocket, as shown in Figure 8, led to interference with the chassis structure.

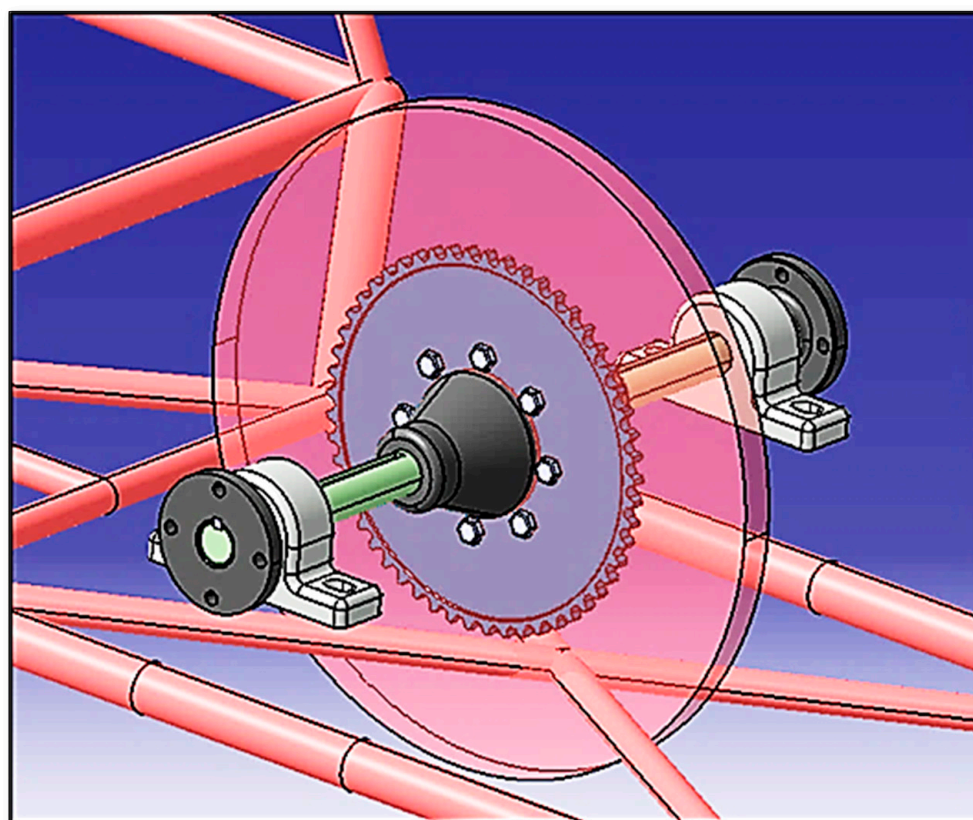


Figure 8. Isometric view of the larger sprocket needed and clashes with chassis.

3. Build Instructions

This section outlines the assembly of the main powertrain components. The design emphasizes modularity, cost-effectiveness, and compliance with Shell Eco-marathon rules.

3.1. Mechanical Assembly Procedure

3.1.1. Motor Mount Installation

The 1500 W BLDC motor was secured to the chassis using a rigid mount with slotted holes for chain tensioning, as shown in Figure 9. This allows adjustability and ensures robustness under load.

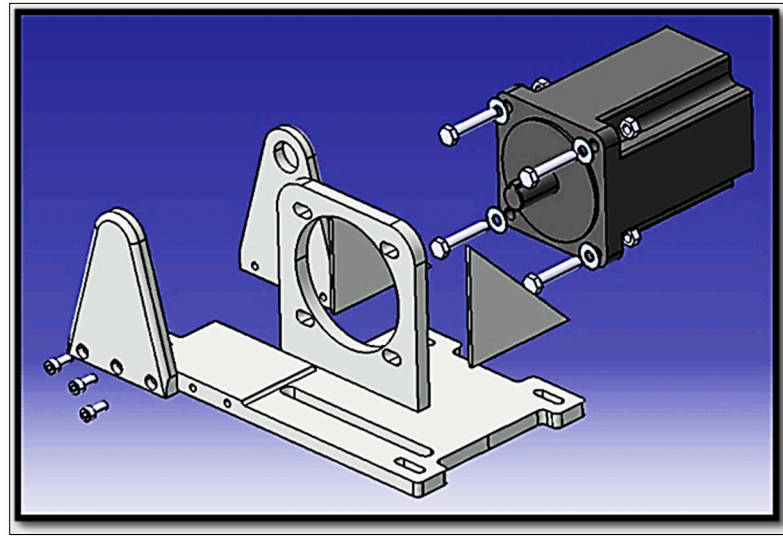


Figure 9. Exploded view of the motor mount.

3.1.2. Two-Stage Chain Drive Setup

A modular bracket system supports the sprockets and intermediate shaft, enabling efficient assembly and straightforward ratio changes, as shown in Figure 10a.

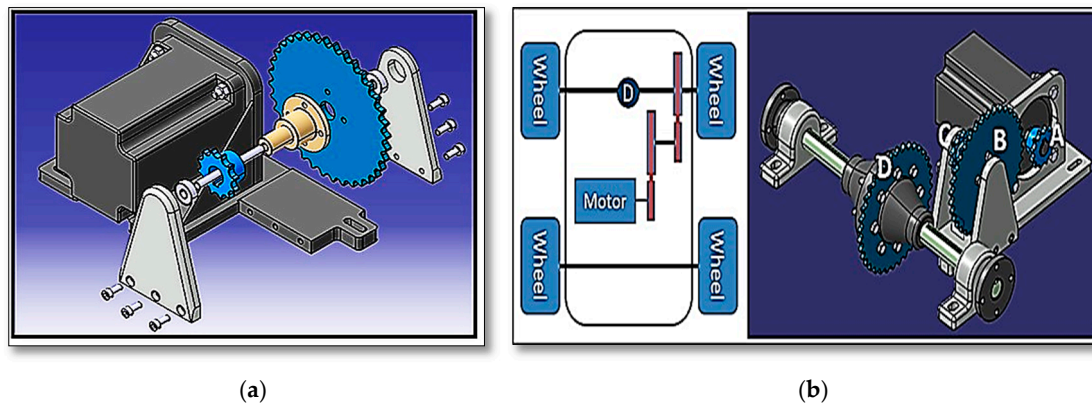


Figure 10. (a) Exploded view of the central transmission assembly. (b) Two-stage chain drive setup.

A two-stage chain drive configuration was adopted, as shown in Figure 10b, allowing for large overall gear ratios while keeping individual sprockets compact enough to fit within the vehicle geometry. The new powertrain design adapted a two-stage chain drive linked through a pin between sprockets B and C.

The benefit of a two-stage chain drive is the possibility of achieving a high gear ratio with the use of smaller-sized sprockets, as the overall gear ratio is the multiplication product of the gear ratios of the two sets of chain drives. The size of the sprockets is listed in Table 9.

Table 9. Required sprocket sizes to achieve desired gear ratio.

Sprocket	Number of Teeth (Gear Ratio = 12)	Number of Teeth (Gear Ratio = 8)
A	12	12
B	32	48
C	12	12
D	36	36

Only one sprocket requires replacement to switch between the 12:1 and 8:1 gear ratios (data summarized in Table 9), simplifying maintenance and tuning.

The final drivetrain design implemented in the vehicle is a two-stage chain drive, as shown in Table 9 and validated by the CAD models in Figures 9 and 10a,b. This configuration allows achieving large gear ratios while accommodating spatial constraints, and it improves modularity and serviceability during gear ratio changes.

3.1.3. Differential Installation

The initial differential choice was the Formula Student limited slip differential from Drexler Automotive, as shown in Figure 11, a company that supplies high-performance vehicle limited slip differentials for motorsports. The Formula Student differential weighs 4.7 kg and is compact in size, but the only concern is the price since the differential alone costs GBP 1500–1900, and this could go up to GBP 3000 for the whole assembly of the differential system.

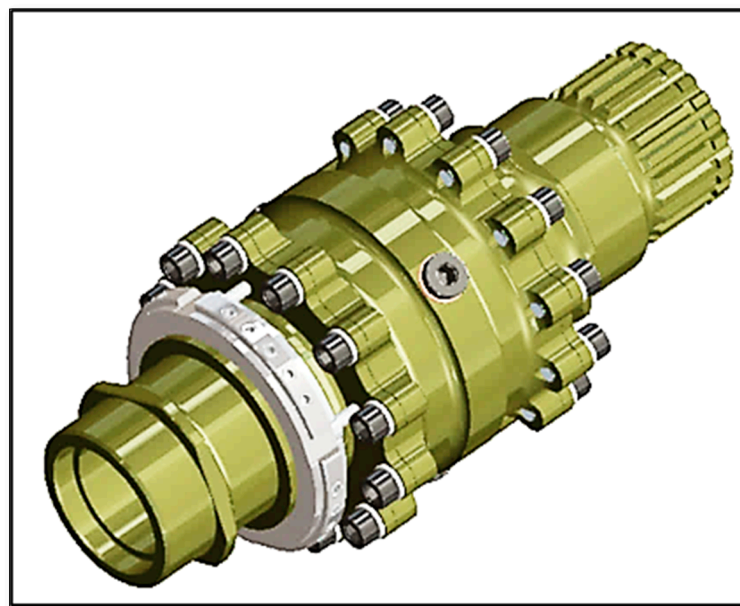


Figure 11. Formula Student LSD.

The alternative is an open differential from Peerless Gear, as shown in Figure 12. The differential is designed for lightweight utility vehicles and go-karts, which is suitable for the Shell Eco-marathon application. It has a more reasonable price of GBP 300 and a weight of 5.5 kg, which is less than 1 kg heavier than the Formula Student differential. Considering that vehicle handling is not the focus of the design and the cost differences between the two options, the team decided to adapt the Peerless differential over the Formula Student differential, as shown in Figure 12.



Figure 12. Peerless 110 differential.

Technical specifications of the Peerless Gear differential:

- Output torque: 305 Nm.
- Weight on differential: 272.3 kg.

The drivetrain uses an off-the-shelf Peerless 110 chain drive differential, commonly applied in go-karts and light utility vehicles. Its use avoided custom design and simplified integration. The supplied 25 mm axles were shortened from 1000 mm to 490 mm to fit the chassis and mounted in bearing housings with slotted holes for adjustability. Each axle includes a 5 mm keyway slot for half-shaft mounts, which were manufactured from aluminium and fitted by interference for secure axial positioning, as seen in Figure 13.

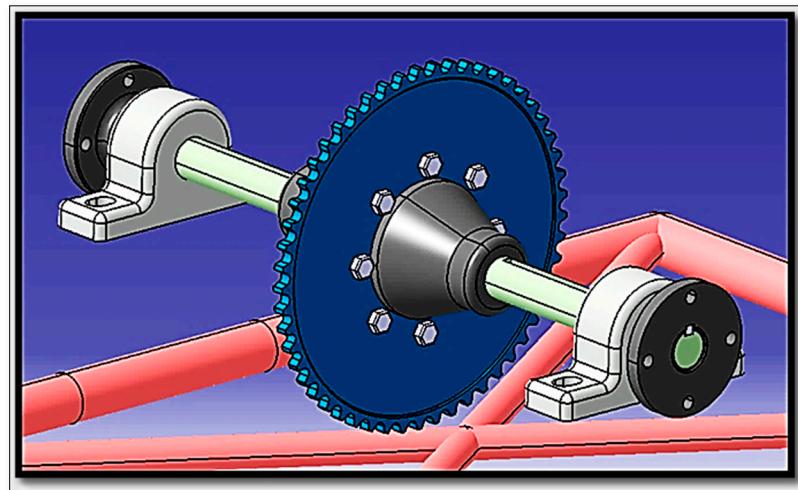


Figure 13. Chain-driven differential assembly.

3.1.4. Modularity, Manufacturability, and Reliability

Several decisions were made to enhance modularity and manufacturability.

Chain tensioning was achieved using the adjustable motor mount slots and the idler pulley system shown in Figure 14, reducing the likelihood of slack-induced failure.

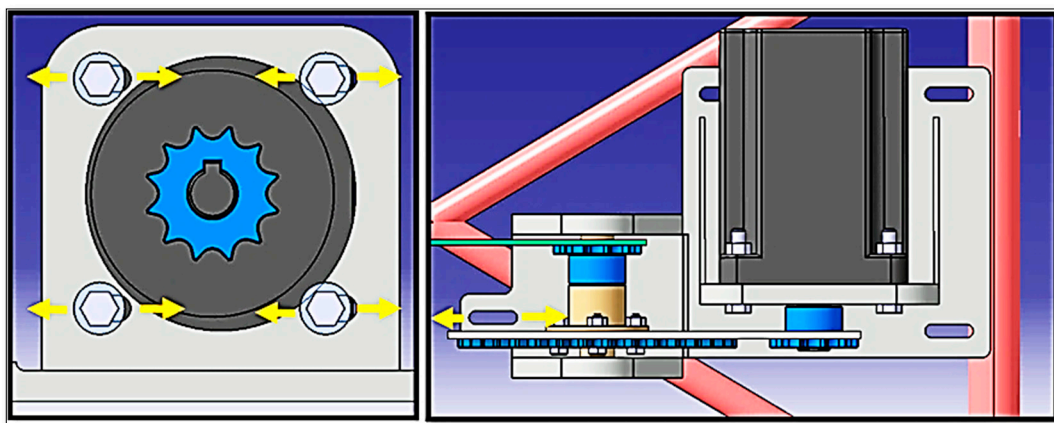


Figure 14. Details of the chain tightening method.

Where possible, off-the-shelf parts were selected to reduce costs, improve ease of replacement, and ensure compatibility with available tools and materials. This structural integration approach offers high reliability, efficient space use, and ease of maintenance, all essential for competition success and compliance with Shell Eco-marathon engineering guidelines.

Electrical Subsystem Assembly

3.1.5. Battery Pack Wiring, BMS, and DC-DC Converter Installation

To follow Shell Eco-marathon 2025 safety regulations, a 12S Li-ion BMS (Daly Electronics Co. Ltd., Shenzhen, China) was integrated inside the battery enclosure. The BMS provides cell balancing and undervoltage and overcurrent protection and automatically isolates the battery under fault conditions.

Shell Eco-marathon official rules:

Vehicle design:

- The vehicle must weigh a maximum of 225 kg, with the driver weighing a minimum of 70 kg.
- The voltage of the system should be 60 V or below.

Battery pack design:

- Maximum battery capacity is 1000 Wh.
- Only one lithium-based battery is allowed.
- Must be equipped with a solid metal containment tray underneath or be enclosed within a battery charging bag.
- Must be fixed outside the driver's compartment behind the bulkhead.
- The charger must be either bought with the battery or purpose-built specifically suited to the battery.

Battery Management System (BMS) design:

- Must be tailored to chemistry to control and protect the battery.
- Must provide cell balancing and overvoltage protection during off-track charging.
- Must automatically isolate the battery (with an added requirement of providing cell-level over-discharge and overcurrent as part of the in-vehicle system).
- Must be in the physical battery package and be powered directly by the battery.

Drivetrain design:

- A maximum of two electric motors can be equipped in the drivetrain.
- Electric motors can be either bought and changed or purpose-built.
- Motor control must be purpose-built.

The battery pack detailed in Figure 15 is mounted in a 3 mm thick aluminium containment tray to meet structural and fire safety guidelines. It is attached to the chassis using four M6 hex bolts, reinforced by plastic 3D-printed straps, and enclosed with a vacuum-formed plastic lid for secure and accessible maintenance.

The battery tray was built from a 3 mm aluminium plate (Al), with the pack secured using M6 bolts and plastic straps under a removable vacuum-formed (VF) plastic lid for easy maintenance. A Victron Orion-Tr 48/12-9A DC-DC (direct current–direct current) converter steps down the 48 V pack voltage to 12 V for auxiliary systems such as the Vehicle Control Unit (VCU), sensors, and lighting. Compact (100 × 113 × 47 mm, 0.42 kg) and 87% efficient, it provides stable, pulse-width modulation (PWM)-adjustable output suitable for competition use [32].

In this design, a unidirectional step-down (buck) DC-DC converter is required to reduce the 48 V battery voltage to 12 V for auxiliary loads (sensors, lights, control units). Adjustable output is useful for components needing 12.5–15.5 V or higher, such as power steering [33].

The selected DC-DC converter offers PWM-adjustable output, parallel connection capability, and reduced current at high temperatures. With 87% efficiency, it minimizes heat loss, while its compact size (100 × 113 × 47 mm, 0.42 kg) makes it well-suited for integration [32].

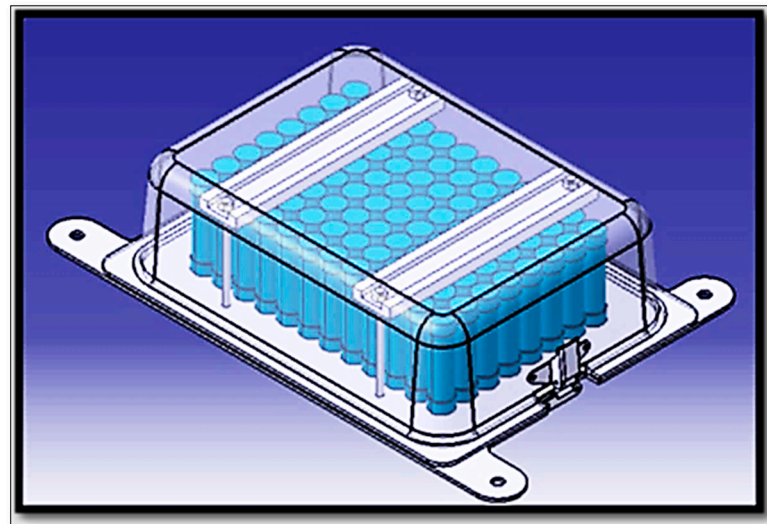


Figure 15. Details of the battery case design.

3.1.6. Motor Controller Wiring Diagram

The custom-built motor controller uses the following:

- MOSFET-based power switching.
- Potentiometer-based throttle input.
- Inputs from the BMS including battery voltage, motor temperature, and current sensing.
- External signals from the brake switch, ignition key, and cooling fan control.

As illustrated in Figure 16 of the report, the controller connects to the motor via U/V/W three-phase outputs. It receives 48 V power input from the battery and 12 V auxiliary input from the DC-DC converter. The controller also routes a PWM signal to the cooling fan, allowing thermal management based on load and ambient conditions. All signal wiring is low-voltage (LV), while power wiring uses 10 AWG (American Wire Gauge) silicone-insulated copper (Cu) cables, chosen for their high current capacity and flexibility during installation.

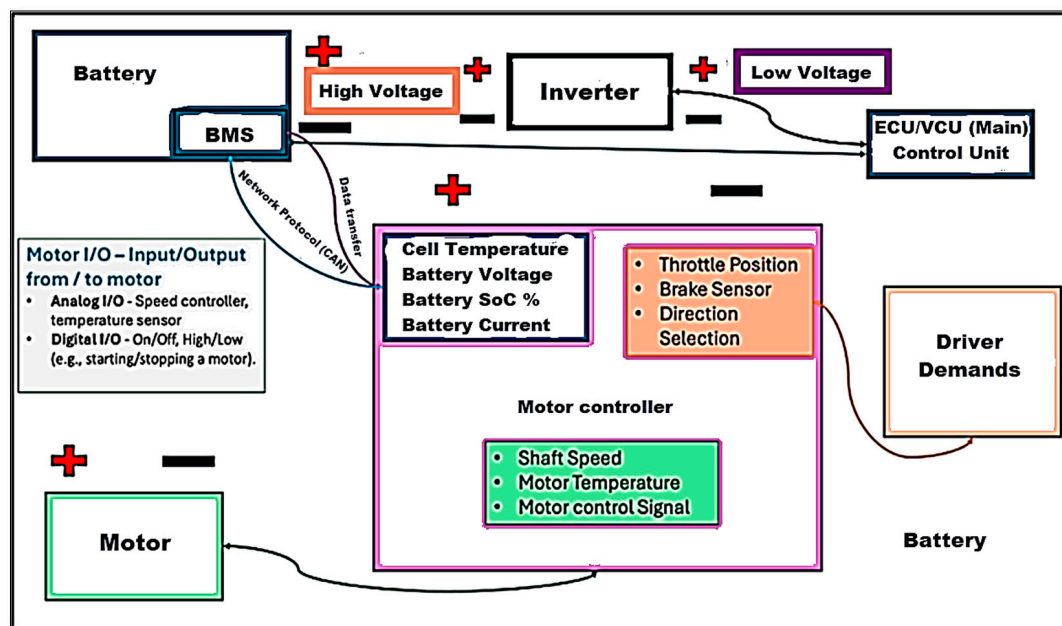


Figure 16. Motor controller diagram.

Motor controllers (MCs) regulate electric motors by controlling start/stop, speed, and torque. They can be manually operated or automatically programmed and either custom-built or purchased off the shelf. In an Urban Concept electric vehicle (EV), the DC motor controller (DCMC) manages motor speed and torque. The high-voltage (HV) side (48 V) powers the motor, while the low-voltage (LV) side (12 V) supplies the DCMC and auxiliary electronics.

The in-house DCMC design primarily uses MOSFETs (Metal–Oxide–Semiconductor Field-Effect Transistors) due to their high efficiency, as they require minimal gate current to maintain switching states, unlike NPN (Negative–Positive–Negative) BJTs (Bipolar Junction Transistors), which rely on base current flow. Supporting components include a potentiometer (POT, variable resistor for voltage control), resistors (R, current limiting/protection), wiring, and a breadboard (BB, prototyping platform) [34].

3.1.7. Connection Schematics for Reverse Gear Logic

To improve campus manoeuvrability, a reverse drive mode was added via a dashboard toggle switch, linked to the controller's reverse-enable pin. Software ensures activation only when stationary, meeting safety rules without added mechanical complexity. The system voltage is limited to ≤ 48 V for safe on-campus charging. The total vehicle budget is GBP 13,500.

4. Operating Instructions

4.1. Power System Initialization

Battery Charging, Voltage Check, and Fuse Configuration

Before powering the vehicle, the custom battery pack must be charged with a charger matched to Molicel P28A Li-ion cells, per Shell Eco-marathon (SEM) rules. The pack's nominal voltage is 43.2 V, below the 60 V limit. The pack voltage must be verified with a multimeter, and all cells must be confirmed to be within safe thresholds via the Battery Management System (BMS). Overcurrent protection is provided by the BMS, with devices rated for the 39 A peak draw. Fuse ratings and continuity must be checked before applying power.

4.2. Drive System Operation

4.2.1. Motor Control via Throttle

Motor control is managed through a PID-based feedback loop implemented in the Simulink model. The driver input from the throttle finds the torque demand, which is processed in real time to support the desired velocity with minimal overshoot. The vehicle followed the drive cycle with minimal overshoot, maintaining control throughout, as shown in Figure 17.

The Electric Drive Unit (EDU) block, illustrated in Figure 18, interprets the torque signal and delivers electrical current to the 1500 W BLDC motor. This system was simplified for simulation using a fixed operating point, but its logic supports stable throttle-to-motor control for smooth acceleration.

4.2.2. Stop/Go Cycle Behaviours and Safe Shutdown

The Shell Eco-Marathon Urban Concept category requires a stop-and-go profile during competition. This is simulated using a drive cycle that includes acceleration, cruising, and stopping phases. The vehicle is optimized to operate at 25 km/h and 40 km/h using gear ratios of 12 and 8, respectively, as shown in Table 4. For safe shutdown, the throttle must first be released fully to bring the vehicle to a stop. Then, the power must be isolated using the system kill switch described in the electronics layout. Although wait times are not

specified, a brief cooldown period is advisable to allow heat dissipation, particularly after extended operation.

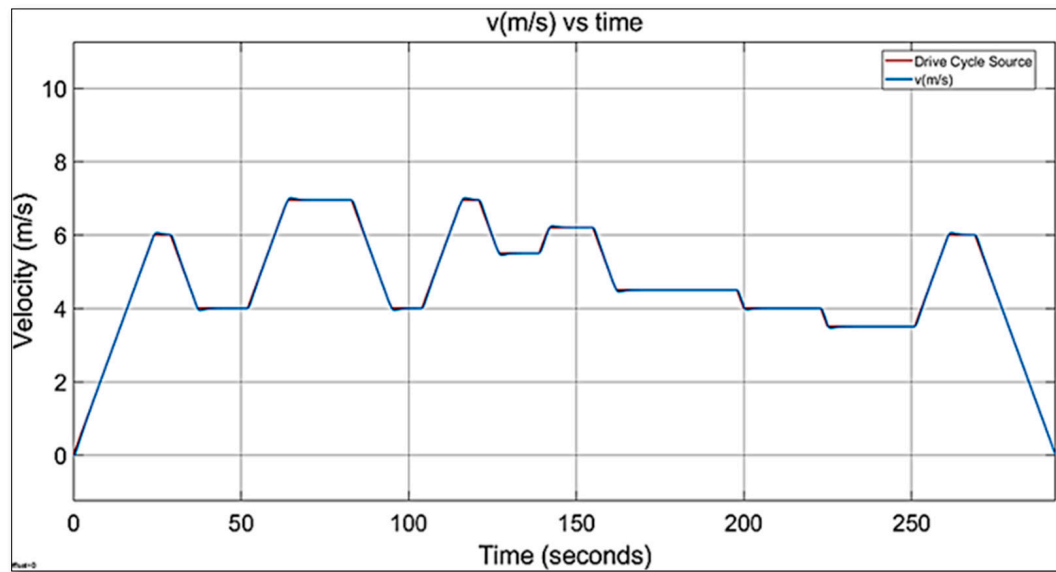


Figure 17. Vehicle speed tracking (km/h vs. time).

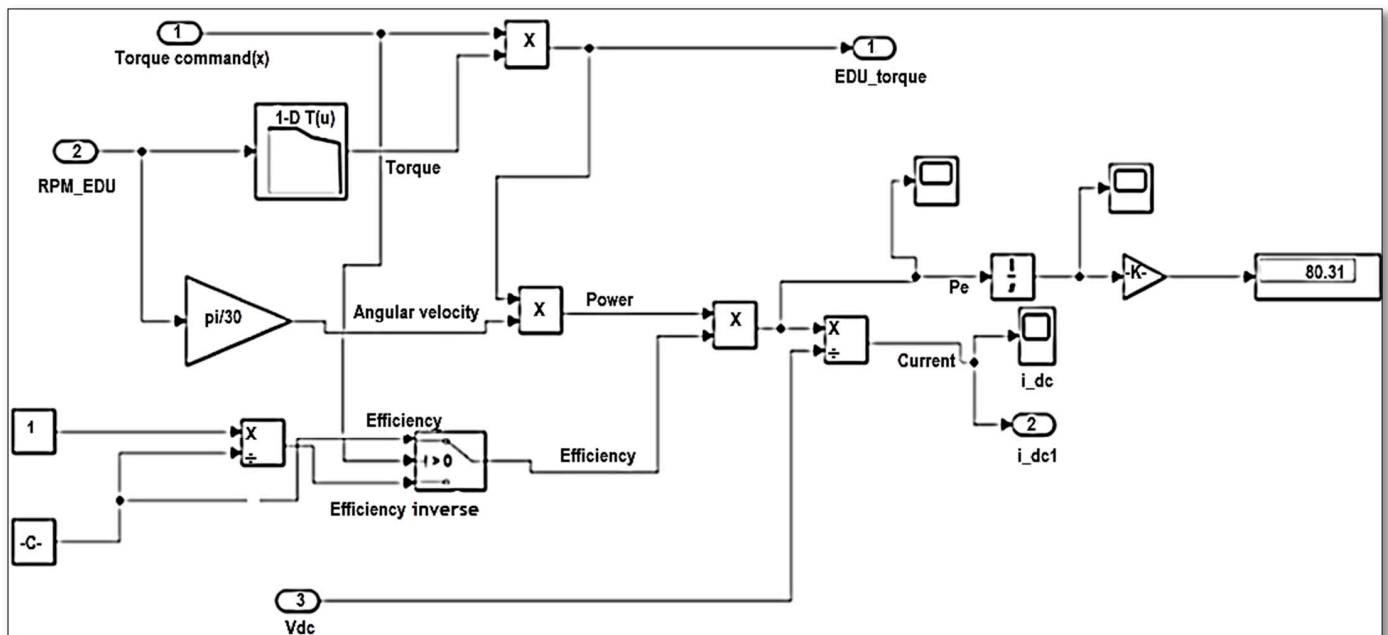


Figure 18. EDU (Electric Drive Unit) subsystem based on 48 V, 1500 W brushless DC motor.

4.3. Safety Measures

Thermal Considerations

The BLDC (brushless DC) motor improves efficiency by eliminating brushes and rotor winding losses, generating less heat under low-speed, low-load Shell Eco-marathon (SEM) conditions, though sustained peak power can still cause heating, requiring unobstructed passive ventilation and temperature checks. The Battery Management System (BMS) ensures electrical protection by monitoring overvoltage, over-discharge, overcurrent, and cell imbalance, with autonomous cutoff with faults. Mechanically, the two-stage chain drive uses a 428 high-strength roller chain (tensile strength 17.8 kN, per BS 228:1994), exceeding

load requirements, to improve durability; reliability further depends on proper tension, alignment, and sprocket wear checks before each run.

5. Validation and Discussion

The drivetrain achieved an energy consumption of 20.95 Wh per lap (equal to 45.8 Wh/km), with the battery State of Charge decreasing only from 100% to 97.8% over a full cycle. This demonstrates the ability to complete multiple laps without recharging.

5.1. Model Overview

The MATLAB/Simulink model evaluates the energy efficiency and control performance of the electric powertrain. The system consists of the following:

- **Drive Cycle Source:** This provides the reference speed for one Shell Eco-marathon lap, derived from track geometry, elevation, and cornering. Uphill sections were modelled with added acceleration, downhill sections with controlled deceleration, and turns with safe cornering speeds. Transitions were smoothed using uniform acceleration/deceleration.
- **PID Controller:** This continuously compares actual and reference speeds, outputting a torque demand to minimize error.
- **Electric Drive Unit (EDU):** This represents a 48 V, 1500 W BLDC motor, converting torque demand into current draw and motor speed. Due to limited torque–speed–voltage data, the EDU was simplified to a fixed operating point using rated motor specifications (48 V, 1500 W, 4.78 Nm at 3000 rpm).

Equations for torque and current at the point of maximum efficiency:

$$Power(kW) = \frac{Torque(Nm) \times RPM}{9549.296}$$

$$Power(kW) = \frac{2.908Nm \times 2991.1RPM}{9549.296}$$

$$Power(kW) = 0.91089316$$

$$Power(W) = Voltage \times Current$$

Battery Subsystem: The battery block simulates voltage, current, and SoC over time using internal resistance and discharge modelling. It is based on a 12S8P configuration of Molicel P28A cells, designed to remain within Shell Eco-marathon limits, as shown in Figure 19.

The battery block used the Molicel P28A 18650 lithium-ion cell and tracked the State of Charge (SoC) by keeping track of how much current flows in or out of the battery over time. When the battery supplies power (discharging), the SoC goes down. When it is charged, the SoC goes up. The SoC was calculated by integrating the negative current over time, scaled by the total capacity of the battery in ampere-hours (Ah). This integration assumes 100% efficiency, meaning all the current going in or out changed the charge level directly, and no losses were made.

The equation used for the SoC is as follows:

$$SoC(t) = SoC_0 - \frac{1}{C_{bat}} \int_0^t i_{dc} dt$$

As shown in the Simulink diagram, the SoC is fed into two lookup tables:

- **Open-Circuit Voltage (OCV),** which provides a voltage estimate based on the SoC.
- **Internal resistance (Rint),** which provides resistance variation across the SoC.

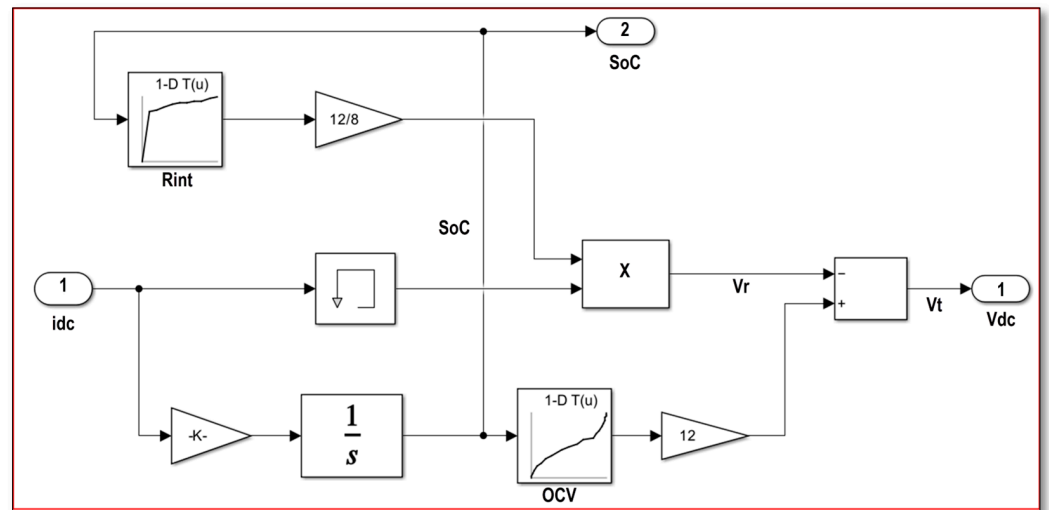


Figure 19. Battery subsystem in the Simulink model, showing a 12S8P configuration of Molicel P28A 18650 lithium-ion cells (43.2 V, 22.4 Ah, 967 Wh).

These values are used to calculate the terminal voltage of the battery using Ohm's Law. The internal voltage drop (V_r) is calculated as follows:

$$V_r = i_{dc} \cdot R_{int}(SoC)$$

Then, the terminal voltage is shown as follows:

$$V_{dc} = OCV(SoC) - V_r$$

This structure enables a more dynamic and realistic voltage output as the battery discharges, with both resistance rise and OCV drop considered. The model does not include any thermal effects or aging but offers a good balance between accuracy and computational simplicity for range estimation purposes.

Vehicle Dynamics Block: This part models longitudinal motion using input torque, factoring in rolling resistance and aerodynamic drag. It outputs the vehicle speed and distance travelled, linking electrical and mechanical subsystems, as shown in Figure 20.

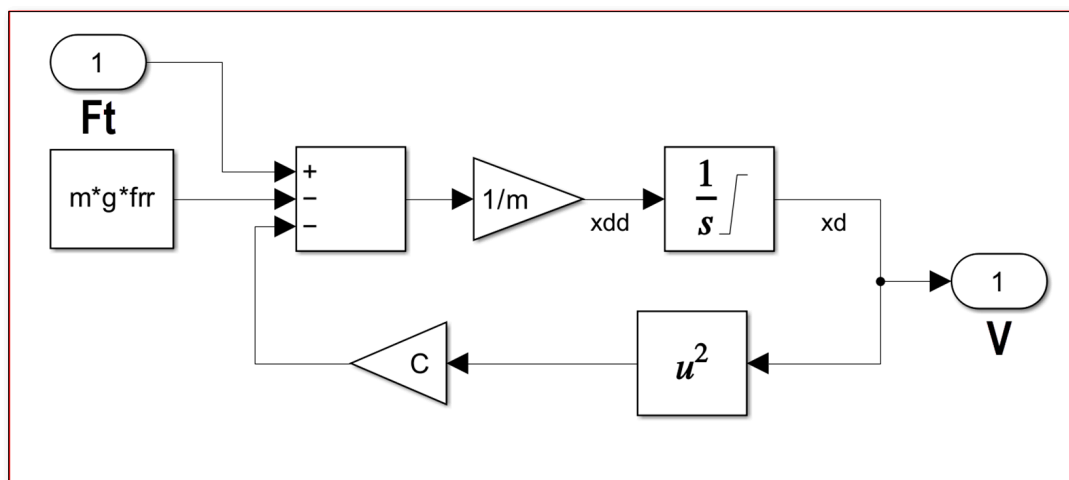


Figure 20. Vehicle Dynamics block in the Simulink model. This converts motor torque into tractive force, accounting for rolling resistance and aerodynamic drag, and outputs vehicle acceleration, velocity, and distance travelled.

The output torque from the EDU was then passed to the Gearbox and Tire block, where it was converted into a tractive force (F_t) using a wheel radius of (r) 0.28575 m and gear ratio of twelve. The forward motion of the vehicle was simulated using a Vehicle Dynamics Block that applied Newton’s second law, calculating the acceleration and velocity from the net force.

The equation used is as follows:

$$F_t = m \times \frac{dv}{dt}$$

$$v(t) = \int a(t)dt$$

To calculate distance, the vehicle speed was passed through an integrator block. The signal created was multiplied by a factor (1/1000) to give the total distance travelled in kilometres. These blocks were implemented modularly to enable independent testing and ensure fidelity before full system integration.

Inputs: These include the drive cycle, terrain, and motor specifications.

Drive Cycle: The simulation uses a custom-developed 293 s drive cycle with varying target speeds, simulating start–stop motion, and moderate acceleration phases typical of Shell Eco-marathon events.

Terrain Considerations: The elevation effects were simplified into acceleration profiles within the drive cycle itself. Added slope data was encoded into the speed reference trajectory to simulate increased load during uphill segments and controlled deceleration downhill.

Motor Specifications: The model uses a 48 V, 1500 W BLDC motor rated at 4.78 Nm and 3000 rpm. The EDU block was developed using mathematical equations representative of this motor’s speed–torque–current behaviour.

5.2. Simulation Results

As recorded in Table 10, the vehicle consumed 20.95 Wh of energy per lap. This value was computed by integrating the instantaneous power draw across the duration of the lap and normalizing it by lap distance. The figure proves that the system meets the Shell Eco-marathon’s stringent energy consumption criteria.

Table 10. Input parameters for Simulink.

Parameter	Value
Total Distance (km)	45.78 km
Total Simulation Time	9788 s
Avg. Speed (km/h)	25 km/h
Max Torque (T_{EDU})	5.9 Nm
Peak Current (i_{dc})	20.066 A
Initial SoC %	100%
Final SoC % After 1 Lap	97.77%
Final Voltage (V_{dc}) After 1 Lap	47.72 V
Energy Consumed After 1 Lap	20.95 Wh

Figure 21 shows that the battery SoC dropped from 100% to approximately 97.77% during one lap. The steady slope of the curve reflects stable current drawing and confirms

the energy consumption result. This outcome confirms that the drivetrain is well within the 1000 Wh battery energy limit and allows for multiple laps without recharging.

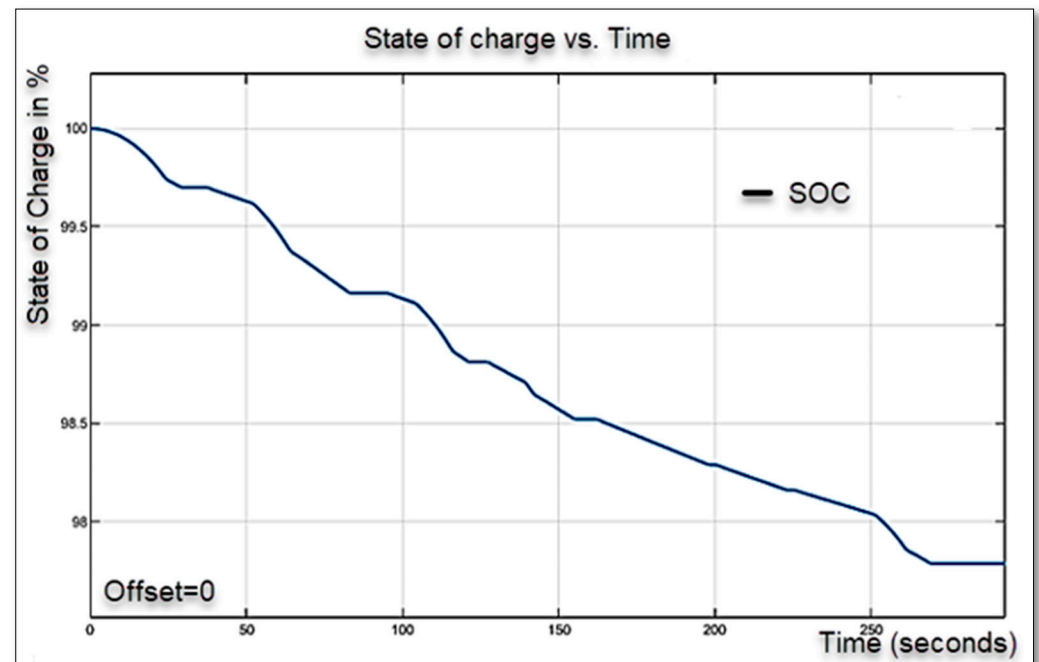


Figure 21. SoC vs. time: this confirms that energy draw was consistent with expected consumption trends.

As illustrated in Figure 22, the actual vehicle speed closely follows the reference profile. This proves the effectiveness of the PID controller in delivering smooth and exact tracking of the target velocity, including during transition phases such as acceleration and braking.

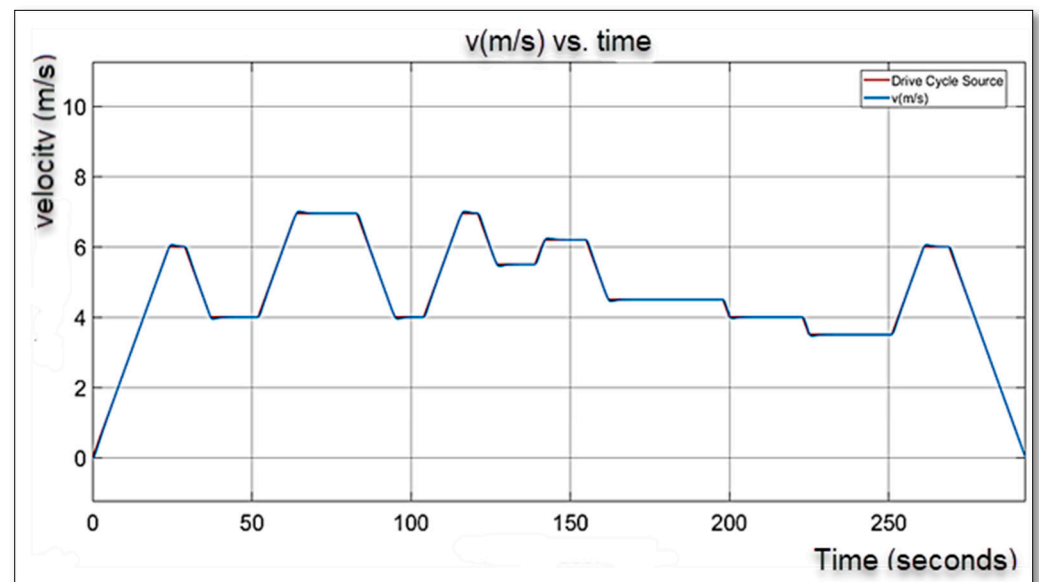


Figure 22. Vehicle speed tracking (km/h vs. time). Validation: This showed a match to the expected efficiency.

5.2.1. Battery Voltage and Current Behaviour

As the internal resistance increased and the SoC decreased, the voltage (V_{dc}) decreased with time. But the graph in Figure 23 displays short spikes and stable voltage periods, which may seem unreasonable at first.

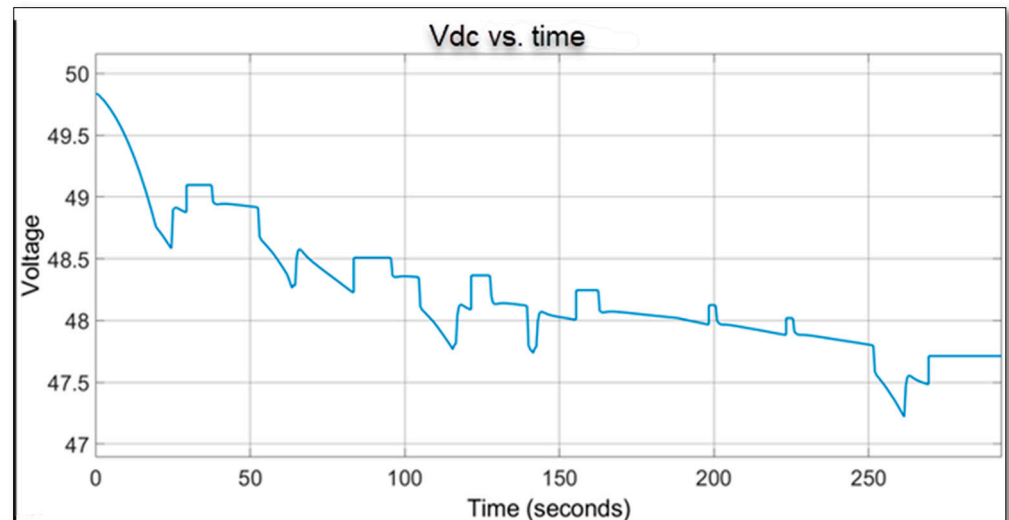


Figure 23. V_dc vs. time.

When the current (i_{dc}) drops, Figure 24, as it does during the driving cycle’s low-speed cruising periods, these brief rises or flat bits happen. The terminal voltage slightly rises when the current decreases because the voltage drops because the internal resistance decreases. This shows the battery’s normal power behaviour and verifies the accuracy of the internal resistance and OCV lookup table model.

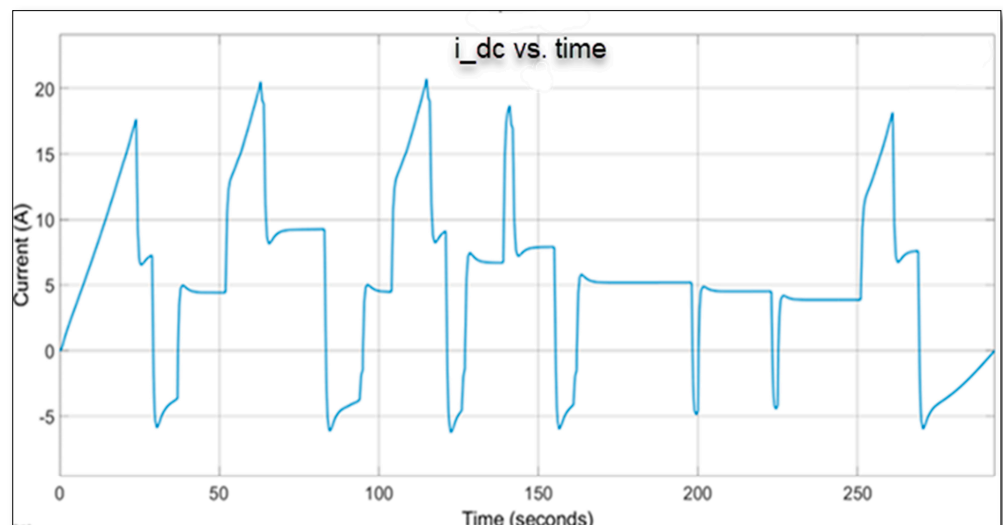


Figure 24. i_{dc} vs. time.

Sharp current peaks are aligned with torque demands and corresponding dips in V_{dc} . Lower i_{dc} values match or slightly increase the voltage. The speed profile confirms that the vehicle tracks the target drive cycle with minimal lag. The voltage remains stable throughout operation, while the current profile reflects an increased draw during acceleration and reduced demand during braking. These trends are consistent with expected electric vehicle behaviour and validate the fidelity of the control strategy. The recorded signals provide the necessary data for power analysis, enabling assessment of power as a function of speed through the following:

$$P(t) = V(t) \times I(t)$$

The battery behaviour was within expectations. The voltage declined steadily overall as the SoC dropped but remained within the limits throughout. The small increases in V_{dc} , rather than being errors, indicated a good model response to the reduced load. These moments of voltage recovery are common in lithium-ion batteries and result from a drop in the internal voltage loss ($V_{dc} = i_{dc} \times R_{int}$) during low-power operation.

The prototype's performance (around 45.8 Wh/km) positions it within the competitive mid-range of SEM Urban Concept vehicles. Top-tier teams such as SZEnergy achieve below 5 Wh/km, while focused vehicles typically range between 20 and 60 Wh/km. Within this context, the present design demonstrates a strong balance of efficiency, safety, and affordability, making it especially suitable for reproducibility in student-led and resource-constrained projects.

5.2.2. Limitations and Future Work

Several simplifications were adopted: regenerative braking, detailed nonlinear motor maps, drivetrain wear, and thermal dynamics were excluded. As such, while the simulation demonstrates strong efficiency potential, experimental testing is required to confirm real-world robustness. Future work will therefore focus on hardware-in-the-loop validation, thermal modelling, and durability testing, which will enhance predictive accuracy and support design improvements.

6. Conclusions

This study presented the design, construction, and simulation-based validation of a high-efficiency electric powertrain for a Shell Eco-marathon Urban Concept vehicle. The proposed system, comprising a 1500 W BLDC motor, custom 12S8P lithium-ion battery pack, and modular two-stage chain drive, achieved an energy consumption approximately equal to 45.8 Wh/km, confirming both regulatory compliance and competitive efficiency.

The novelty of this work lies in combining low-cost, modular hardware with a validated MATLAB/Simulink framework, producing a reproducible and transferable drivetrain solution. This approach supports not only SEM teams but also wider applications in micro-mobility and scalability to real-world micro-EVs.

When benchmarked against other SEM Urban Concept vehicles, the prototype falls within the competitive mid-range of efficiency. For example, the TIM UPS-INSA team achieved 251.3 km/kWh (around 3.98 Wh/km) in 2025 [35], while the EVA team reported 47.6 km/kWh (around 21.0 Wh/km) [36]. These results demonstrate a broad performance range in the battery electric Urban Concept category.

Although not at the extreme upper end of performance, the present design balances efficiency with cost-effectiveness, manufacturability, and safety, key factors for reproducibility in academic and resource-constrained environments.

Future work will extend validation beyond simulation to include thermal and durability testing, drivetrain wear analysis, and hardware-in-the-loop experiments, ensuring predictive accuracy and robustness under real-world operating conditions.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/designs9050113/s1>. CAD: Cad File (Catia V5), The full CAD modelling assembly design of the drivetrain and battery system; Ev model with drive cycle: Simulink model (.slx), Shell Eco-marathon 2025 prototype electric vehicle Simulink model; 6053MAA_Innovation_Group_Industrial_Report (.pdf), Automotive product innovation industrial report; BoM_EV_Powertrain: Excel Worksheet (.xlsx), Bill of materials for powertrain design; Efficiency-Oriented_Drive_Cycle_Map_NEW: Excel Worksheet (.xlsx), Efficiency-oriented drive cycle map data (time–speed).

Author Contributions: K.H., major contributions: conceptualization; methodology; data curation; writing—original draft preparation; and writing—review and editing. S.E., major contributions: conceptualization; methodology; visualization; supervision; project administration; and resources. A.D.C.: critique of assumptions and limitations. J.J., S.H.L., S.S. and R.Y., major contributions: methodology; software; validation; formal analysis; and investigation. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: Data and materials are available upon reasonable request.

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